

Collision-Free Ball Navigation in MuJoCo: PPO, RRT Planning, and Feedback Control

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Part I

Collision-Free PPO Navigation

1 Introduction

The current system combines model-free reinforcement learning with geometric planning and feedback control. A clipped proximal policy optimization (PPO) agent produces continuous forces for a ball in MuJoCo, while a rapidly-exploring random tree (RRT) supplies collision-free waypoints. Before PPO training, the actor imitates a proportional-derivative (PD) path-following expert. During training and evaluation, a safety shield blends the learned action with the expert action and rejects predicted geometry violations. This hybrid design addresses a limitation of the original direct-goal actor-critic implementation: a distance-only policy can converge toward the near side of the central obstacle instead of learning to travel around it.

The state is

$$s_t = (x_t, y_t, \dot{x}_t, \dot{y}_t),$$

and each policy observation augments the state with the displacement to the current RRT waypoint,

$$o_t = (x_t, y_t, \dot{x}_t, \dot{y}_t, x_t^{\text{wp}} - x_t, y_t^{\text{wp}} - y_t).$$

The action is a two-dimensional force vector. MuJoCo advances the physical state using a simulation time step of 0.1 seconds.

2 Environment

The workspace and obstacle geometry are read from the MuJoCo model rather than duplicated as training constants. The goal is fixed at $(0.8, 0.0)$, and an episode succeeds when the ball is within $\epsilon = 0.1$ of that point. Starts are sampled uniformly from collision-free space, at least 2ϵ from the goal. Spawn and planning clearances account for the ball radius; the RRT uses a 0.06 clearance around walls.

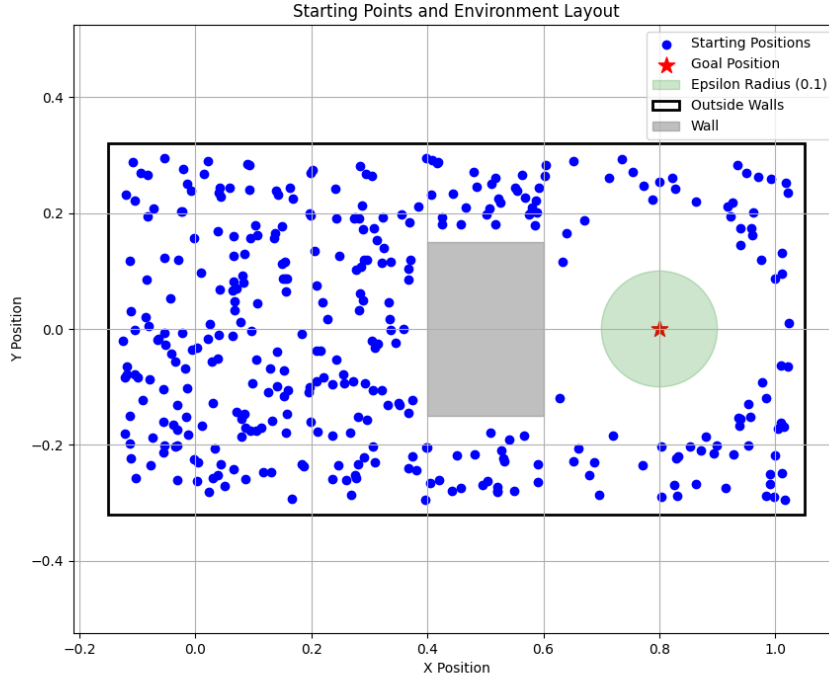


Figure 1: Random collision-free training starts, obstacle geometry, and goal region.

3 Hybrid Planning and Control

3.1 RRT Waypoints

For every training or evaluation episode, an RRT is planned from the sampled start to the goal. The planner uses a step size of 0.10, a goal-sampling probability of 0.20, and at most 1000 iterations. Candidate points and segments are checked against the outer boundary and an inflated central obstacle. The resulting path is shortcut-smoothed and then densified so adjacent targets are no more than 0.10 apart. If the direct start-to-goal segment is collision-free, it is densified without growing a tree.

3.2 PD Demonstrations

The feedback expert generates bounded actions

$$a_t^{\text{PD}} = \tanh(K_p(p_t^{\text{wp}} - p_t) - K_d\dot{p}_t), \quad K_p = 6, \quad K_d = 6.$$

Before PPO begins, the actor is trained for 30 epochs on 4096 states sampled along collision-free RRT segments. This imitation stage gives the policy an initial obstacle-aware path-following behavior instead of requiring it to discover basic control entirely through sparse successes.

3.3 Safety Shield

The deployed control blends the policy and expert outputs:

$$a_t^{\text{safe}} = 0.25a_t^\pi + 0.75a_t^{\text{PD}}.$$

A one-step position prediction is checked against the environment with a 0.025 geometry clearance. If the prediction is unsafe, the shield substitutes the expert action. The final force is clipped component-wise to $[-1, 1]$. The shield is used consistently during data collection, deterministic evaluation, and rendering.

4 PPO Actor–Critic Model

The actor and critic are separate fully connected networks with hidden widths 32, 64, and 32, ReLU activations, and Xavier initialization. The actor outputs the mean and standard deviation of a Gaussian distribution. The standard deviation is produced with a softplus function and constrained to $[0.05, 0.5]$. A sampled raw action u_t is transformed with tanh before it is sent to the safety shield. The critic estimates the scalar value $V(o_t)$.

PPO uses the probability ratio

$$r_t(\theta) = \frac{\pi_\theta(u_t | o_t)}{\pi_{\theta_{\text{old}}}(u_t | o_t)}$$

and the clipped actor objective

$$L_{\text{actor}} = -\mathbb{E}_t \left[\min \left(r_t \hat{A}_t, \text{clip}(r_t, 1 - \varepsilon_c, 1 + \varepsilon_c) \hat{A}_t \right) \right],$$

where $\varepsilon_c = 0.2$. Advantages are computed with generalized advantage estimation using $\gamma = 0.99$ and $\lambda = 0.95$, then normalized. The critic uses smooth L_1 loss. The combined optimization objective includes a value coefficient of 0.5, entropy coefficient of 0.01, and gradient clipping at 0.5.

5 Training

Training runs for 360 episodes in batches of 64, with up to 1800 MuJoCo steps per episode. Each collected batch is optimized for six PPO epochs using minibatches of 1024 samples and an Adam learning rate of 3×10^{-4} .

For a transition to next state s_{t+1} , the implemented shaped reward is

$$r_t = -2d_{\text{goal}}(s_{t+1}) - 0.01 + 10(d_{\text{wp}}(s_t) - d_{\text{wp}}(s_{t+1})) - 2d_{\text{cross}}(s_{t+1}) + 100\mathbb{I}_{\text{goal}} - 25\mathbb{I}_{\text{collision}}.$$

The terms penalize goal distance and elapsed time, reward progress toward the current waypoint, penalize cross-track error from the active path segment, reward goal entry, and strongly penalize collision. Episodes terminate on success, collision, or timeout.

After every training batch, the mean policy is evaluated without exploration on 32 newly sampled episodes. Checkpoint selection prioritizes lower collision rate, then higher success rate, and finally higher mean reward. Rendering loads this best checkpoint and uses deterministic mean actions.

6 Results and Discussion

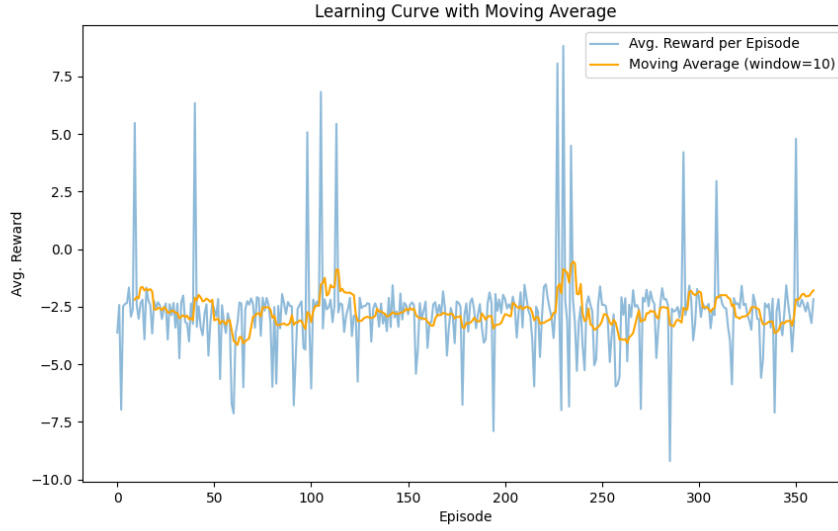


Figure 2: Average training reward per episode and a ten-episode moving average over 360 episodes.

The reward trace is noisy because episodes begin at different distances from the goal and PPO samples stochastic actions. Positive spikes correspond to successful episodes receiving the terminal bonus. The moving average remains mostly negative, so this plot alone does not demonstrate monotonic convergence; deterministic success and collision rates are the more meaningful checkpoint criteria.

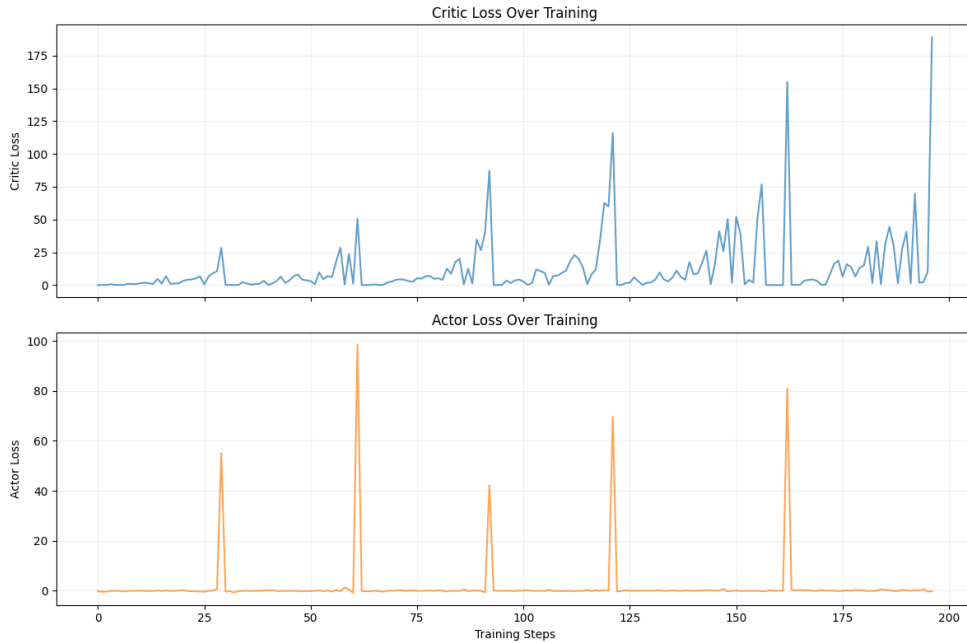


Figure 3: Critic and clipped PPO actor losses during training.

Most actor updates remain near zero because clipping limits large policy changes, with

occasional spikes from batches containing unusually large normalized advantages. The critic also shows intermittent large errors associated with terminal rewards and variable episode outcomes. These traces support the use of gradient clipping and deterministic evaluation, but they should not be interpreted as proof of collision-free generalization.

The best saved checkpoint was selected after 64 training episodes. In its deterministic evaluation over 32 newly sampled episodes, it reached the goal in all 32 episodes and recorded no collisions. Its mean per-step reward was 0.6235. Table 1 summarizes the checkpoint metadata stored with the model.

Metric	Best checkpoint
Training episodes completed	64
Deterministic evaluation episodes	32
Success rate	100%
Collision rate	0%
Mean reward per step	0.6235

Table 1: Deterministic evaluation metrics stored in the best PPO checkpoint.

7 Conclusion

The revised implementation is a hybrid collision-avoidance system rather than a standalone actor-critic agent. RRT provides route-level obstacle avoidance, PD imitation initializes and stabilizes waypoint tracking, PPO improves the stochastic policy, and the safety shield constrains executed actions. The separation of training, planning, environment, rendering, and plotting interfaces also makes the system easier to evaluate and extend. Future experiments should compare the learned policy against the expert-only controller, report confidence intervals across random seeds, and ablate RRT guidance, imitation pretraining, and the safety shield.

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Part II

Kinodynamic RRT with PID Control

8 Introduction

In robotics, autonomous path planning and control are crucial aspects of robot navigation, particularly in environments with obstacles. The objective of this project is to implement a hybrid approach that combines kinodynamic rapidly-exploring random Tree (RRT) for path planning and proportional-integral-derivative (PID) controllers for executing the motion along the planned path. This work is demonstrated using a ball in a MuJoCo simulation environment, where the ball navigates through an environment containing walls and boundaries. The goal is to move the ball from a starting position to a goal position while avoiding obstacles using the kinodynamic RRT algorithm and accurately controlling the movement with PID.

8.1 Objectives

The main challenge is efficiently generating a collision-free path from a start point to a goal point in a cluttered environment and ensuring the ball follows this path with minimal deviation. We undertook the following steps to achieve this objective:

- Configuring MuJoCo and exploring various models, including Newton’s cradle with no gravity.
- Developing basic pathfinding in a 3D environment using Dijkstra’s algorithm for initial path selection and fixed velocity controls.

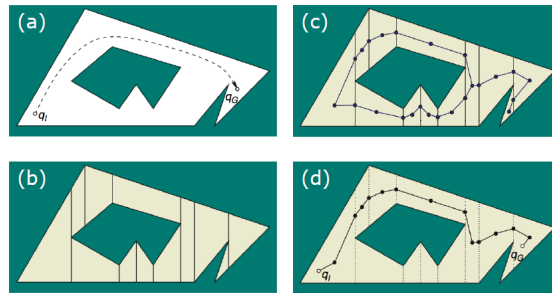


Figure 4: Trapezoidal decomposition ($\mathcal{C} = \mathbb{R}_{\max}^3$)[1]

- Enhancing pathfinding with kinodynamic RRT, incorporating a safety margin around obstacles.
- Replacing basic controls with a PID controller to follow the planned path.
- Simulating and visualizing the robot’s movement within the MuJoCo physics engine.

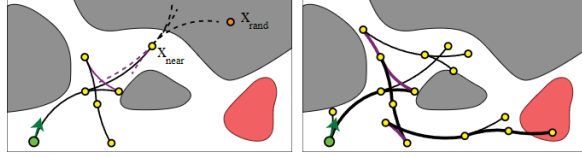


Figure 5: Kinodynamic RRT

9 Environment

The environment consists of a 2D plane with predefined walls and boundaries. The simulation is conducted using the MuJoCo physics engine, which allows for the creation and control of dynamic models. For this project, a ball is moved through the environment containing a set of obstacles to reach the goal. The model used is defined in the `ball_square.xml` file, and the walls are represented as both external boundaries and internal obstacles.

9.1 Map

- **Dimensions:** The 2D environment spans from $x = -0.5$ to $x = 1.5$ and $y = -0.4$ to $y = 0.4$.
- **Obstacles:** A rectangular obstacle is placed in the middle of the environment between coordinates $(0.5, -0.15)$ and $(0.6, 0.15)$, acting as a wall.
- **Goal Area:** The target area for the robot is set between coordinates $(0.9, -0.3)$ and $(1.1, 0.3)$.
- **Boundaries:** The outer walls of the environment prevent the robot from moving outside the area.

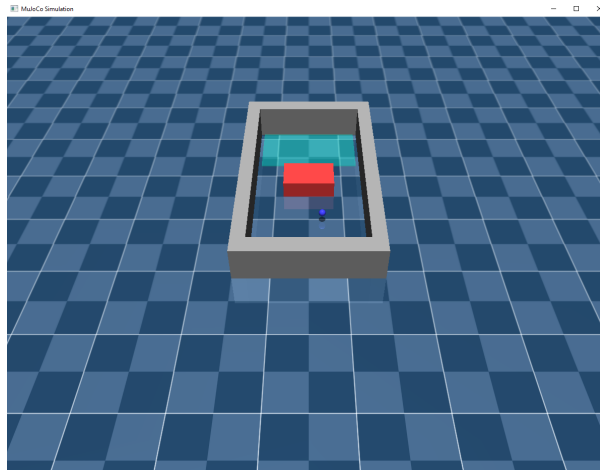


Figure 6: Environment Setup

This setup ensures the presence of both static and dynamic constraints, making path planning non-trivial and control challenging.

10 Path Planning

10.1 Algorithm Overview

RRT[2] is widely used for motion planning in high-dimensional spaces. In this project, we implement a kinodynamic variant of RRT. Unlike standard RRT, the kinodynamic RRT accounts for the robot's dynamics while generating the tree, ensuring that the planned path can be followed realistically by the robot. This approach helps avoid sharp or unrealistic turns that the robot may not be able to follow.

The algorithm grows a tree from the start position by:

1. Sampling a random point in the environment.
2. Identifying the nearest node in the tree.
3. Applying a control input to move the robot towards the sampled point, simulating its dynamics.
4. Checking the resulting position for collisions.
5. Adding the new position to the tree if no collision occurs.

This process repeats until the robot reaches the goal area or a maximum number of iterations is reached. The function `kinodynamic_rrt` implements this process, while `simulate` handles the forward simulation of the robot's dynamics.

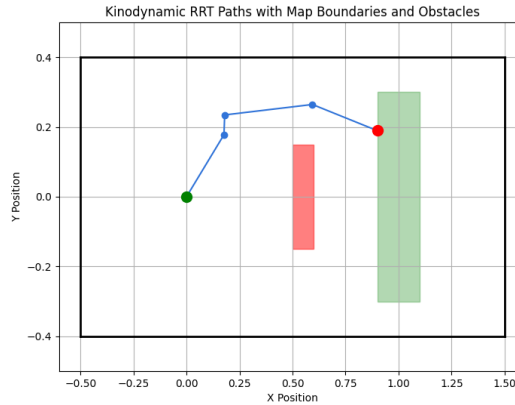


Figure 7: Kinodynamic RRT

10.2 Purpose of `simulate()` in `kinodynamic_rrt`

In traditional RRT, paths are generated based solely on geometric feasibility. However, for real systems with dynamic constraints, `simulate()` serves several critical functions:

- **Ensures Dynamic Feasibility:** Calculates movement towards each sampled point, adjusting with PID control.
- **Incremental Adjustment:** Makes small, corrective adjustments to follow a feasible path.

- **Collision-Free Verification:** Confirms both geometric and dynamic feasibility.
- **Realistic Path Execution:** Validates that each RRT segment can be executed under real-world constraints.

10.3 Collision Avoidance

Collision detection is handled by the function `is_collision_free`, which checks if the robot's new position lies within any of the obstacle regions or the environment boundaries. A safety margin is applied to ensure that the robot does not approach obstacles too closely. The function `is_collision_free_line` checks if a straight line between two points is free of obstacles by verifying that there are no collisions with specified walls.

10.4 Path Construction

Once the algorithm successfully identifies a path to the goal, the tree is traversed from the goal node back to the start node to construct the complete path. This backtracking process is handled by the function `construct_path`. The final path is then visualized, displaying the trajectory, environmental boundaries, and any obstacles encountered along the way.

When a path is established between two nodes, the function `line_goes_through_goal` checks if the path intersects with the goal area by interpolating and sampling points along the path. If an intersection is detected, the endpoint is adjusted to the coordinates (0.9, y) based on the last generated point. This modification effectively minimizes the distance required to reach the goal, as evidenced by the time trial paths.

10.5 Smoothing

After selecting the path within the Rapidly-exploring Random Tree, post-processing occurs using the `smooth_path` function. This function randomly selects points in the graph to determine if they can be directly connected while eliminating intermediate points. As a result, the path is reduced to a series of critical waypoints, simplifying navigation. This reduction allows the control implementation to restart upon reaching each subsequent point, providing the controller with adequate time to complete the feedback loop before moving on to the next waypoint.

11 Navigation

11.1 Controller

PID controllers are commonly used in control systems to achieve stable and accurate control. In this project, two PID controllers are implemented: one for controlling the movement of the ball in the x -direction and the other for the y -direction. Each controller computes the control input based on the error between the current and target positions, aiming to reduce this error over time.

The function `move_ball_to_position_with_pid` implements the control loop, where the ball's current position is compared to the target position from the planned path. The

PID controllers compute the necessary force to move the ball toward the target, and the control inputs are applied in each iteration of the simulation.

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{d}{dt} e(t) \quad (1)$$

where:

- K_p : Proportional gain,
- K_i : Integral gain,
- K_d : Derivative gain,
- $e(t) = r(t) - y(t)$: Error at time t .

11.2 Parameters

The gains for the PID controllers were manually tuned to achieve smooth movement without overshooting. This tuning required observing the patterns in the model simulation and adjusting accordingly; no penalty for previous errors was applied. The proportional (**kp**), integral (**ki**), and derivative (**kd**) gains were set as follows:

- **Proportional Gain (kp)**: 0.58—provides a basic correction based on the current error.
- **Integral Gain (ki)**: 0—not used in this implementation to avoid cumulative errors from drift.
- **Derivative Gain (kd)**: 0.5—helps to smooth the response by anticipating future errors based on the rate of change.

The higher proportional gain results in more aggressive error correction, which was necessary to achieve trials below a certain time threshold. Once the new point is set, the ball minimizes the error by updating its velocity. The x -direction and y -direction have the same controller configuration for symmetry. The same controllers are used to simulate the motion planning between two nodes in the Rapidly-exploring Random Tree.

11.3 Pathfinder

The PID controllers move the ball sequentially through each point in the planned path. Movement is clamped to prevent overshooting, and the simulation continues until the ball reaches the goal position with minimal error (set to 0.075 in this case).

The deviation is calculated using the `calculate_deviation` function. This feedback is important for determining how far the ball is from the trajectory, what adjustments to the controllers are necessary, and ensuring that the ball will not collide with obstacles between two points.

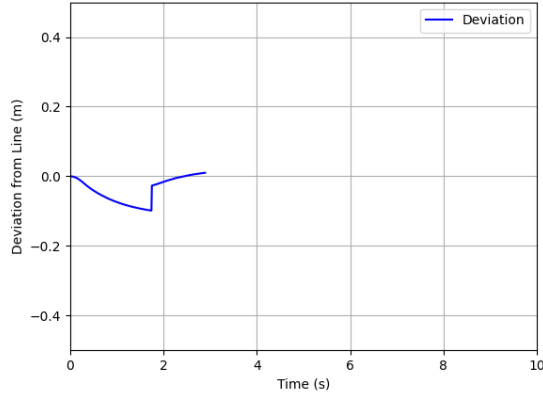


Figure 8: Path Deviation

11.4 Adjustments

Several challenges arise during the control process, including handling sharp turns in the path and adjusting the control parameters to prevent oscillations. The derivative gain plays a crucial role in damping the response and ensuring smooth transitions between points. The best method for finding the right parameter was increasing the K_p value until oscillation was guaranteed. The Ziegler–Nichols tuning method [4] is a heuristic approach to tuning a PID controller, developed by John G. Ziegler and Nathaniel B. Nichols. It is used to determine the proportional, integral, and derivative gains K_p , K_i , and K_d for a PID controller.

Controller Type	K_p	K_i	K_d
P	$0.5K_u$	-	-
PI	$0.45K_u$	$\frac{K_p}{0.83T_u}$	-
PD	$0.8K_u$	-	$K_p \cdot 0.125T_u$
PID	$0.6K_u$	$\frac{K_p}{0.5T_u}$	$K_p \cdot 0.125T_u$

Table 2: Ziegler–Nichols Tuning Parameters

11.5 Path Visualization and Obstacle Mapping

The function `visualize_final_tree` generates a graphical representation of the kinodynamic RRT, illustrating the exploration process and highlighting the path to the goal. The easiest approach to path visualization was created a top down 2D view for visualizing the path on the map. This perspective aided the team in debugging how the path was getting created, the optimal path, and dangers in specific areas. As an example, the 2d view showed that the ball was getting too close to the front of the obstacle. Based on the limited controls and the time constraints, the ball would get too close and not know how to get out of the situation. The new information allowed use to implement a buffer around the object preventing the path from entering that area.

11.6 Dynamic Speed Adjustment for Smooth Deceleration

The function `update_speed` adjusts the robot's speed based on its distance from the goal, allowing for controlled deceleration to avoid overshooting.

12 Results

There were three main objectives in this experiment. The first one was getting the environment running, the second generating a path using kinodynamic RRT, and the third was getting the ball to the goal using controllers. The team demonstrated success on all three objectives.

12.1 Environment

The model creation and execution worked well in the MuJoCo environment and allowed the team to witness the controller behavior prior to using in the kinodynamic RRT search. The simulated search for the path should reflect accuracy how the ball would actually move in the space to the way points and maintain control. It prevented the team was implement controller that behave erratically and would not lead to generating a complete path.

12.2 Visualize Trees

At the end of the report, a complete tree generated using kinodynamic RRT used as a path to the goal. The trees show the effectiveness of using kinodynamic RRT and the success rate for finding the goal despite the different seed. In trails 2, 4, and 5, the tree is not expansive showing the quickness for finding a node in the goal. An area worth exploring in the future is implement a suite of controllers and randomly selecting them for a better representation. In this experiment, the team is using one controller for all the path finding.

12.3 Time Trials

The `run_planning_trial` function runs multiple trials and evaluates the success rate of the kinodynamic RRT in reaching the goal area in the a defined time limit. The team was successful in getting a path each trail and under the time constraints with a 100% accuracy. The quickest shows the effectiveness for using a PID controller for path finding and simulating the ball movements to better represent the actual movement. Similar to the drone movement in the University of Utah technical report [3], kinodynamic RRT is a reasonable approach for solving this problem.

13 Conclusion

This project successfully integrated kinodynamic RRT for path planning and PID control for path execution in a simulated 3D environment. By leveraging these techniques, ball can navigate a complex environment with obstacles and boundaries while accurately following the planned path to the goal.

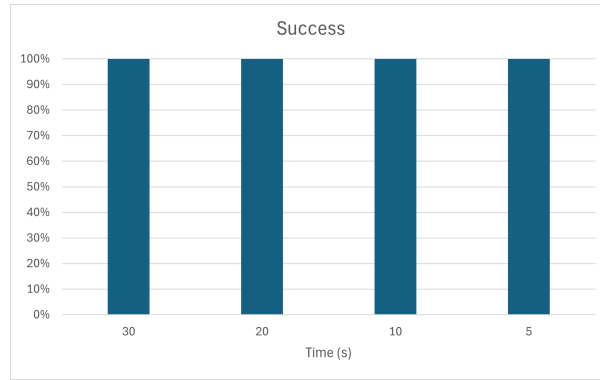
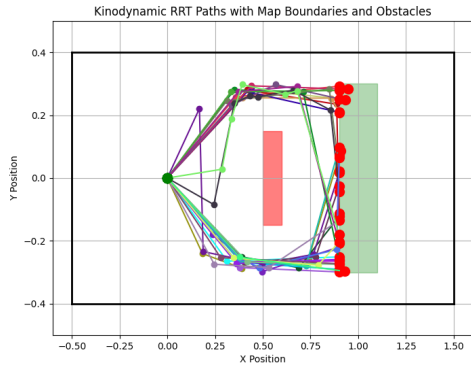


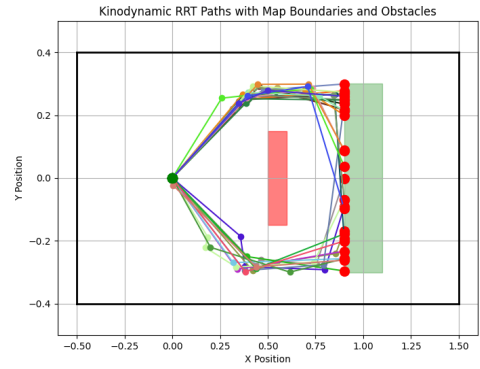
Figure 9: Success Rates

References

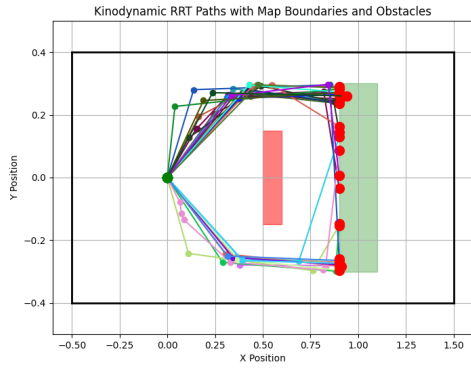
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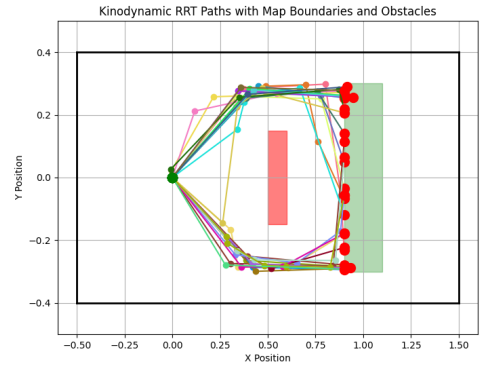
(a) $T_{\max} = 5$



(b) $T_{\max} = 10$

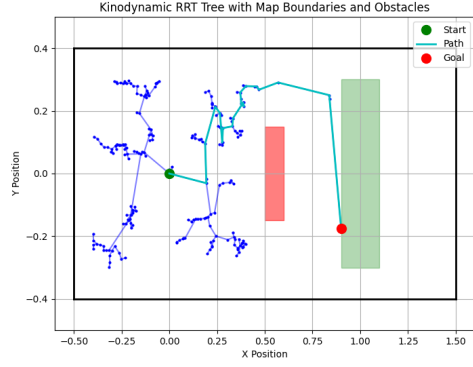


(c) $T_{\max} = 20$

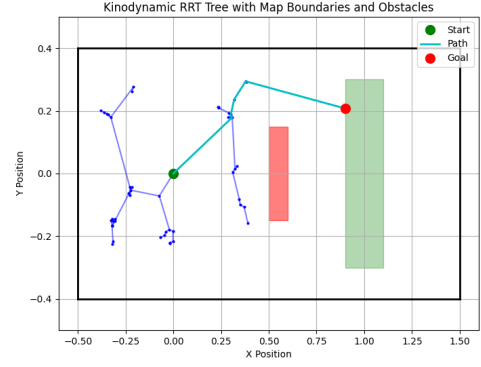


(d) $T_{\max} = 30$

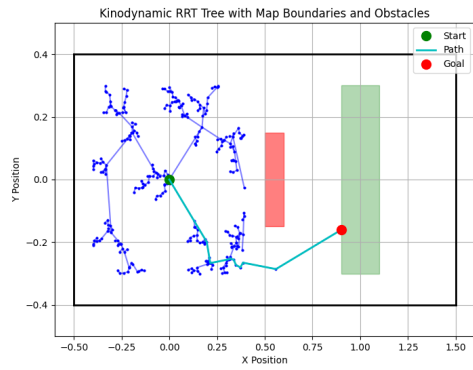
Figure 10: Visualization of path exploration



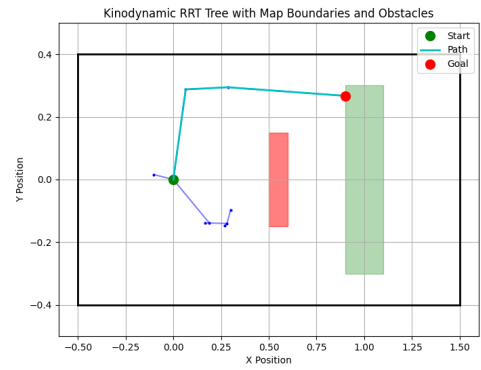
(a) Trial 1



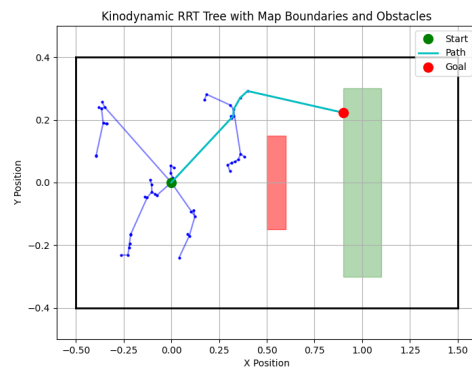
(b) Trial 2



(c) Trial 3



(d) Trial 4



(e) Trial 5

Figure 11: Visualization of trees for different trials.

Part III

3D Fluid Navigation Extension

14 Scope of the Extension

The original collision-free PPO implementation described in the preceding part was extended from planar navigation to controlled motion through a three-dimensional fluid volume. This part records the extension without replacing the earlier baseline. The current system adds a volumetric incompressible-flow solver, buoyancy, turbulence, two-way fluid–structure interaction, three-axis motor control, unrestricted 3D RRT planning, and a single submerged goal.

The updated state and action are

$$s_t = (x_t, y_t, z_t, \dot{x}_t, \dot{y}_t, \dot{z}_t), \quad a_t = (F_{x,t}, F_{y,t}, F_{z,t}).$$

The policy observation contains the six physical state variables and the 3D displacement to the active waypoint, giving nine inputs. The actor produces three normalized motor commands. The horizontal motors provide up to 15 N in each direction, while the vertical motor provides up to 80 N so that the robot can descend against buoyancy. Controller updates occur every 0.02 s, with two 0.01 s MuJoCo physics substeps per control update.

15 Fluid Environment

15.1 Tank and Goal Geometry

The visible water is a translucent, non-colliding MuJoCo box. It fills the tank from $z = 0$ to $z = 0.15$, above the central obstacle top at $z = 0.10$ and below the boundary-wall tops at $z = 0.20$. The only navigation goal is a sphere centered at

$$g = (0.9, -0.22, 0.06),$$

with a strict 3D success radius of 0.01 m. There is no surface goal. The application reports the final 3D position and Euclidean distance when it declares success.

Training starts are sampled throughout the safe water depth. The tank floor is $z = 0$, while the ball center is sampled from $z = 0.02$ to $z = 0.15$ to respect its 0.02 m radius. Samples inside the central obstacle volume are rejected; samples above the obstacle are valid. Figure 12 shows the resulting volumetric distribution.

3D Starting Positions and Underwater Goal

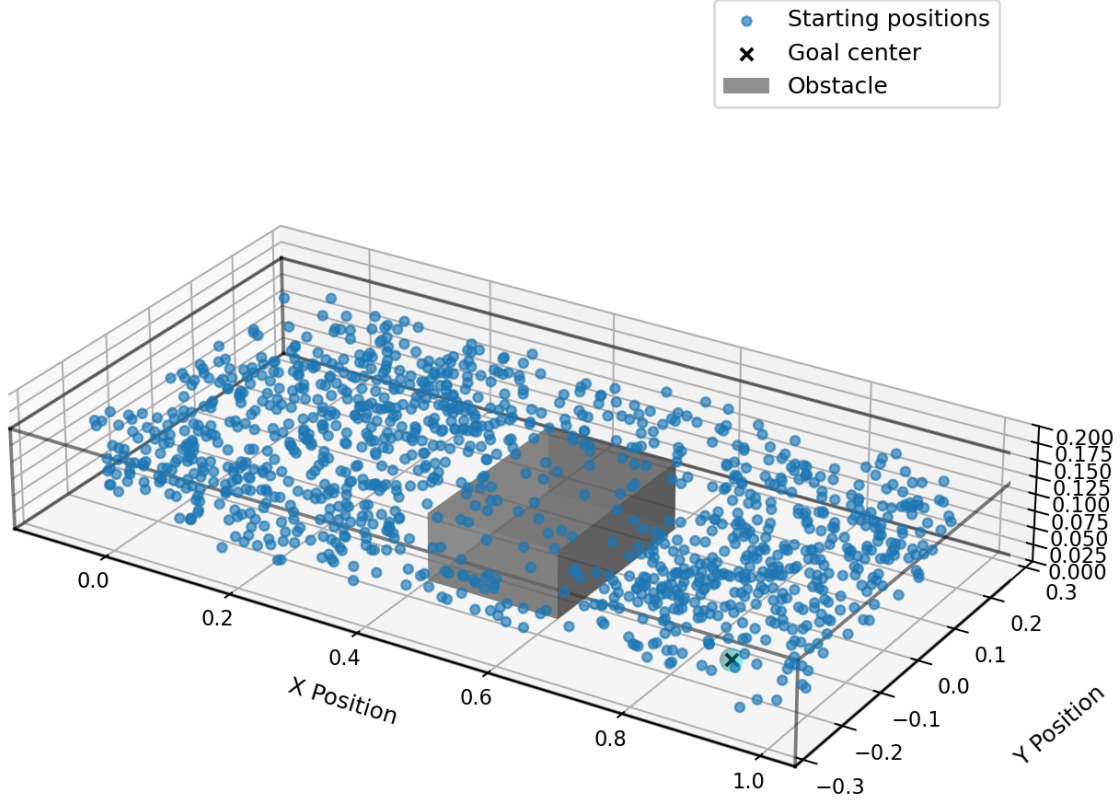


Figure 12: Three-dimensional training starts, tank bounds, obstacle volume, and the single underwater goal.

15.2 Three-Dimensional Navier–Stokes Solver

The water state is discretized on a $32 \times 18 \times 10$ Eulerian grid. Each cell stores pressure p and velocity $\mathbf{u} = (u, v, w)$. The solver advances the incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0.$$

Water density is $\rho = 997 \text{ kg/m}^3$, dynamic viscosity is $\mu = 8.9 \times 10^{-4} \text{ Pa s}$, and kinematic viscosity is $\nu = 8.93 \times 10^{-7} \text{ m}^2/\text{s}$. A semi-Lagrangian backtrace performs advection, Jacobi iterations apply implicit viscous diffusion, and a pressure Poisson solve projects the velocity onto a divergence-free field. The tank floor and central obstacle use no-slip conditions. The upper grid boundary is a free surface with zero normal flow and a zero normal gradient for the tangential velocity. Inlet and outlet boundaries follow the selected current direction. Arrow keys rotate that direction gradually at $60^\circ/\text{s}$ while preserving current speed.

15.3 Turbulence

Large-scale, time-varying, divergence-free eddies inject turbulent energy into all three velocity components. Semi-Lagrangian numerical damping is offset by vorticity confinement,

$$\mathbf{f}_\omega = \epsilon \frac{\nabla \|\boldsymbol{\omega}\|}{\|\nabla \|\boldsymbol{\omega}\|\|} \times \boldsymbol{\omega}, \quad \boldsymbol{\omega} = \nabla \times \mathbf{u}.$$

After 400 solver steps, the measured mean speed is 0.055 m/s, the 95th-percentile speed is 0.141 m/s, and the maximum is 0.261 m/s. The 95th-percentile vorticity magnitude is 4.97 s^{-1} , with a maximum of 11.81 s^{-1} . Relative to the 0.025 m/s inlet current, this represents medium-to-strong turbulence. Figure 13 plots velocity direction and magnitude with arrows; arrow color represents vorticity magnitude.

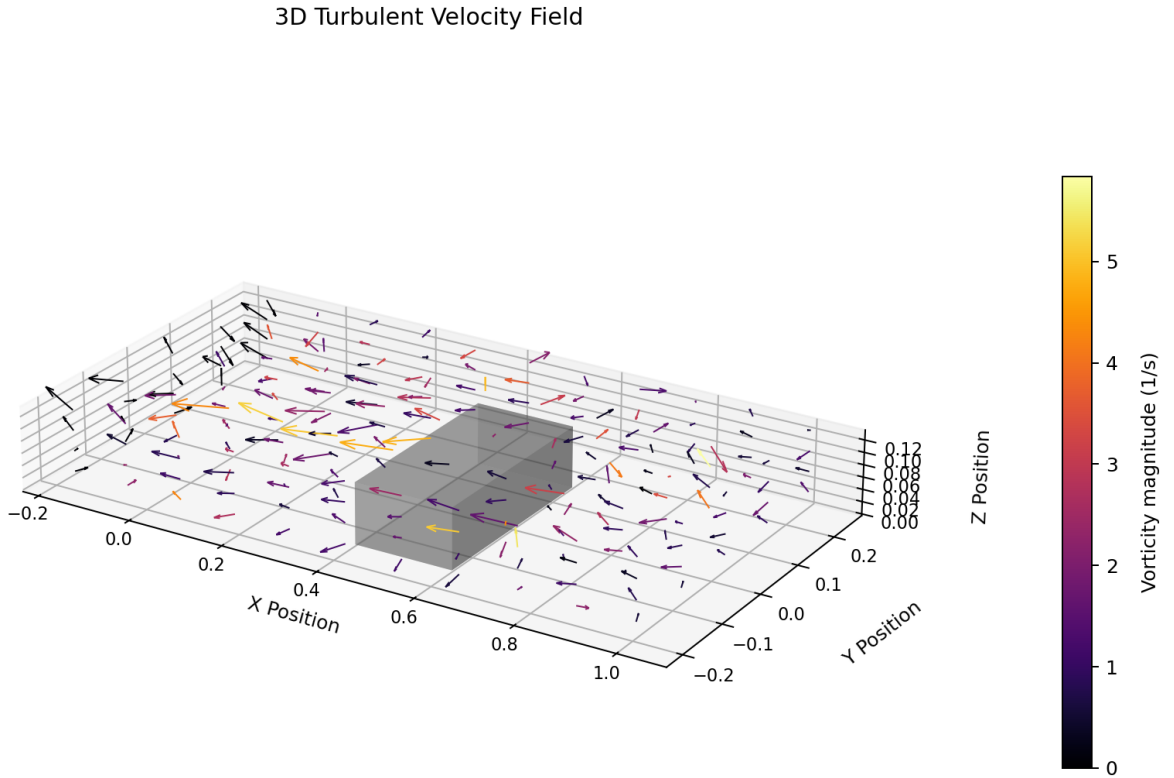


Figure 13: Developed 3D turbulent velocity field. Arrow direction and length represent local velocity; color represents vorticity magnitude in s^{-1} .

16 Hydrodynamic Forces and Fluid–Structure Interaction

The dominant hydrodynamic forces on the spherical robot are evaluated from the local flow around its surface. Relative velocity $\mathbf{v}_r = \mathbf{v}_{\text{robot}} - \mathbf{u}$ produces quadratic form drag

and low-speed Stokes drag,

$$\mathbf{F}_D = -\frac{1}{2}\rho C_D A \|\mathbf{v}_r\| \mathbf{v}_r - 6\pi\mu r \mathbf{v}_r, \quad C_D = 0.47, quad A = \pi r^2.$$

An added-mass term opposes acceleration of the surrounding displaced water, using an added-mass coefficient of 0.5.

For a sphere intersecting the free surface, submerged volume is evaluated as a spherical cap,

$$V_{\text{sub}} = \pi h^2 \left(r - \frac{h}{3} \right),$$

where h is clipped to $[0, 2r]$. Archimedes' force is

$$F_B = \rho g V_{\text{sub}}.$$

The simulated robot mass is much larger than the literal displaced mass of its small visual sphere, so a calibrated displacement model is used: half submersion balances robot weight, full submersion provides upward restoring force, and depth-dependent motor feed-forward supplies exactly the force needed to hold each underwater waypoint. This avoids the previous behavior in which a constant downward bias skipped shallow waypoints or pinned the robot to the tank floor.

Coupling is two-way. The ball is rasterized into the Eulerian grid as a moving no-slip immersed boundary. The fluid applies drag and added-mass forces to the ball, and the equal-and-opposite impulse is distributed to nearby fluid cells, creating a wake. The static obstacle, tank floor, and walls exchange momentum through their fixed no-slip constraints.

17 Three-Dimensional Planning and Control

The current RRT samples unrestricted (x, y, z) configurations throughout the water volume. It uses a step size of 0.075, goal bias of 0.15, and up to 2500 iterations. Collision tests include the obstacle's height, ball radius, wall clearance, floor, and water surface. Consequently, a path may go above the obstacle, around it, descend before it, or combine horizontal and vertical motion; underwater commands are not forced into a surface traversal followed by a vertical dive.

Initial planning retries RRT three times for the original sampled start. It does not resample the start and does not substitute an unchecked straight-line path. During execution, each episode tracks its best goal distance. If no meaningful improvement occurs for 250 control steps (5 simulated seconds), the planner constructs a new collision-free RRT route from the robot's current 3D position and the PID expert temporarily takes full control. This online replanning and recovery mechanism addresses persistent waypoint stalls without changing the episode's initial-state distribution.

Figure 14 shows five representative plans from random depths. The paths demonstrate that the planner uses the full volume: depending on the sampled start and tree growth, it can pass above or around the obstacle while changing depth progressively toward the underwater goal.

Example 3D RRT Paths

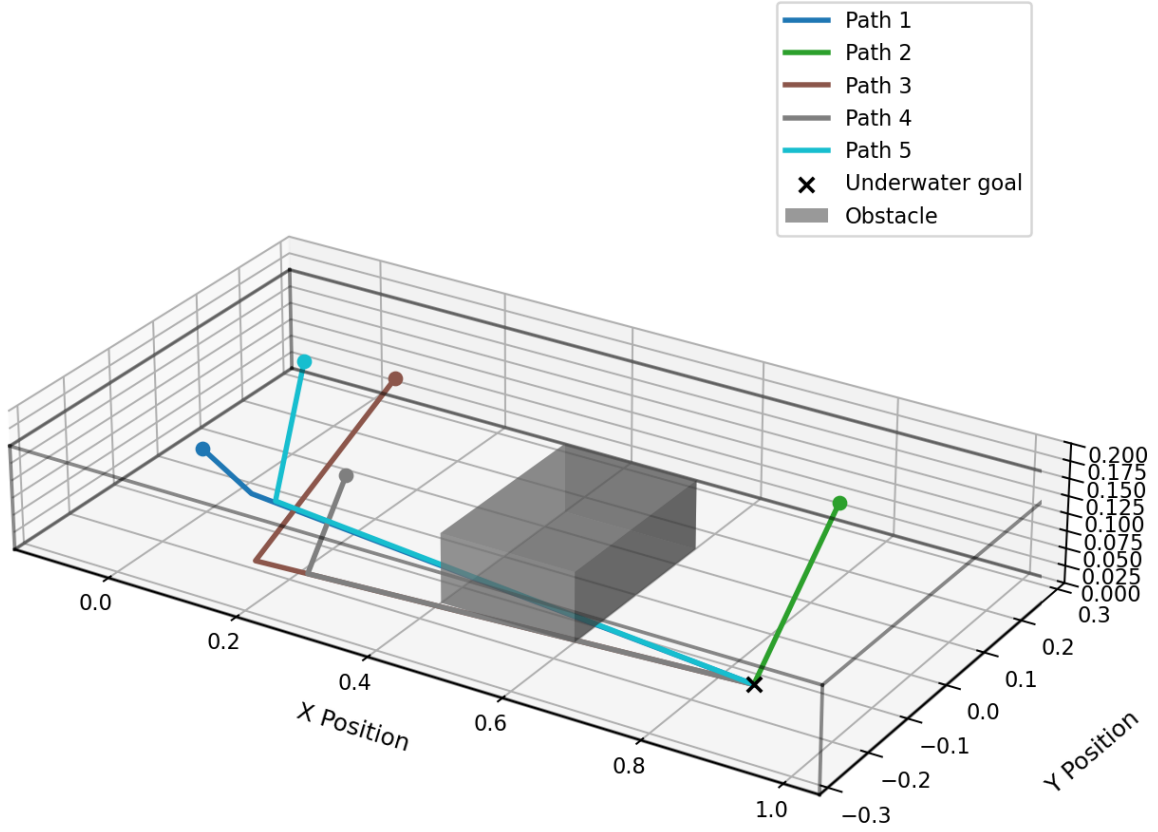


Figure 14: Five collision-free 3D RRT paths from random safe starting depths to the single underwater goal.

The 3D PD expert uses horizontal position and velocity feedback. Its vertical component is expressed in force units and includes hydrostatic feed-forward computed for the target waypoint depth. PPO actions are accepted only when their dot product with the expert action is positive. An aligned learned action receives at most 5% authority; a disagreeing action receives none. Unsafe one-step predictions trigger velocity braking. This alignment-aware shield is intentionally conservative because deterministic collision avoidance and accurate underwater arrival are higher priorities than policy autonomy.

18 Updated Reward and Training

The earlier absolute-distance reward produced a mostly negative, flat trace because distance cost dominated small per-step progress and the terminal bonus was diluted by

per-step averaging. The updated reward is progress-dominant:

$$\begin{aligned} r_t = & 25(d_g(s_t) - d_g(s_{t+1})) + 10(d_{wp}(s_t) - d_{wp}(s_{t+1})) \\ & - 0.02d_g(s_{t+1}) - 0.002 - 0.5d_{cross}(s_{t+1}) - 0.002\|a_t\|^2 \\ & - 0.05\mathbb{I}_{\text{stagnant}} + 25\mathbb{I}_{\text{goal}} - 25\mathbb{I}_{\text{collision}}. \end{aligned}$$

The learning chart now reports total episode return rather than mean reward per step. Training uses 1200 episodes, random starts at all safe depths, 30 epochs of imitation pre-training, batches of 64, six PPO epochs per batch, and a maximum of 4800 control steps per episode, corresponding to 96 seconds of simulated time. PPO rollout collection uses the low-cost deterministic approximation of buoyancy, quadratic and viscous drag, and the mean current. The higher-resolution Navier–Stokes field remains active during interactive rendering; separating the two keeps large batched training runs computationally practical.

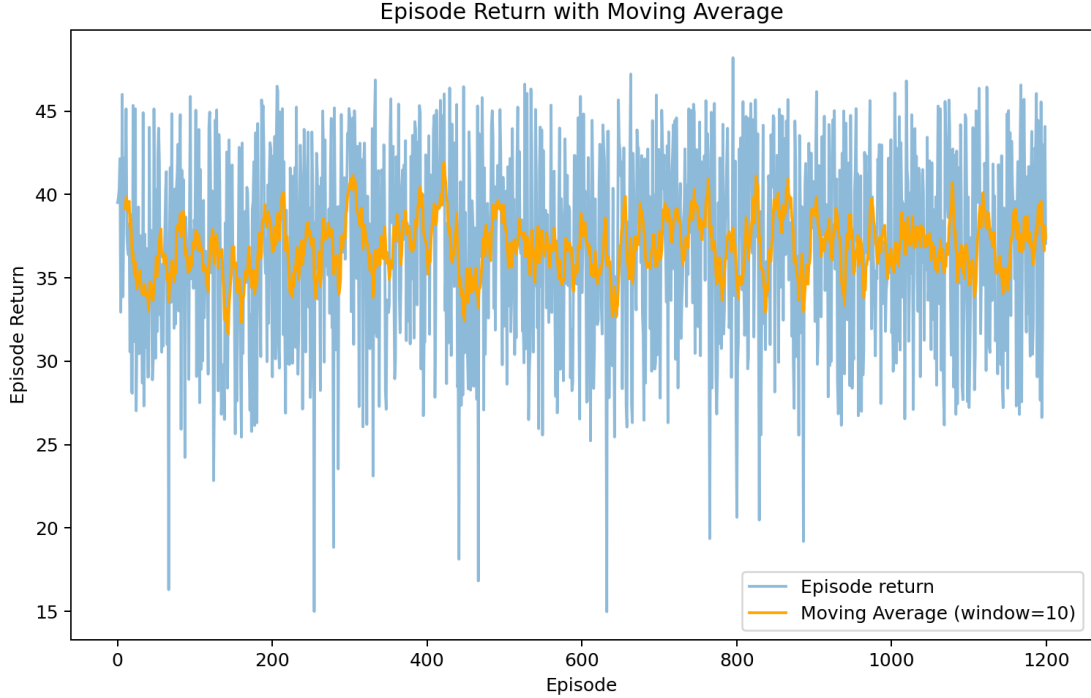


Figure 15: Episode return and ten-episode moving average for depth-randomized underwater training.

All 1200 training episodes reached the underwater goal without collision or timeout. Episode returns remain positive, with most values between approximately 25 and 48; the former large negative timeout spikes are absent. The moving average remains broadly stable because imitation pretraining and expert shielding already provide high success at the beginning of PPO training. Further reward shaping is not presently justified; a stronger test of policy learning would instead reduce expert authority gradually.

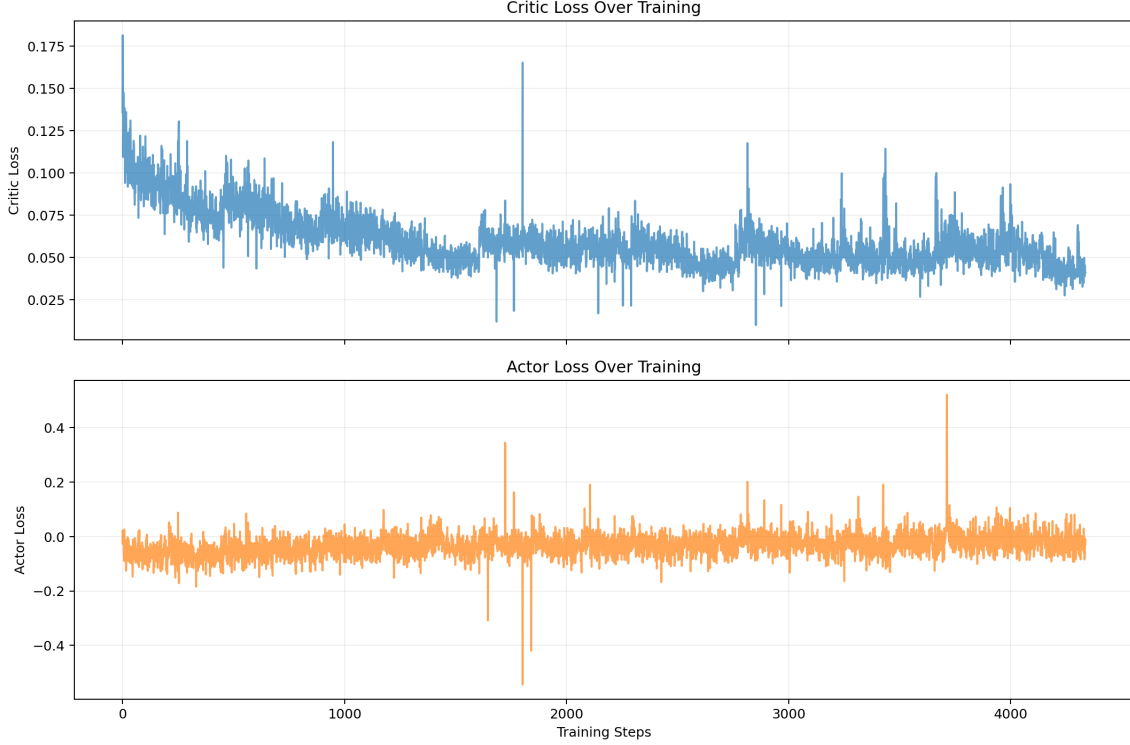


Figure 16: Critic and clipped PPO actor losses for the updated reward.

19 Current Results

The best checkpoint was selected after 704 episodes. Its stored evaluation over 32 randomized-depth episodes reported 100% success, 0% collisions, and mean per-step reward 0.1091. No evaluation episode timed out. Table 3 summarizes the current result.

Metric	Current 3D fluid checkpoint
Training episodes	1200
Selected checkpoint episode	704
Stored evaluation success	100%
Collision rate	0%
Timeout rate	0%
Stored mean reward per step	0.1091
Training goal radius	0.015 m

Table 3: Performance of the depth-randomized underwater controller.

20 Fluid Extension Conclusion

The extension converts the original planar navigation problem into a three-dimensional underwater-control problem. The system now combines a 3D Navier–Stokes approximation, medium-to-strong turbulence, calibrated buoyancy, two-way immersed-boundary coupling, volumetric RRT, three-axis motors, and a strict submerged target. Random-depth training and independent evaluation show that the conservative hybrid controller

maintains high success without recorded wall collisions. Remaining work should quantify performance across multiple random seeds, compare the full solver with laminar and one-way-coupled ablations, and reduce expert authority gradually to measure the learned policy’s independent contribution.

Part IV

Evolutionary Ecosystem Extension

21 Motivation and Scope

The latest extension turns the tank from a single-robot navigation benchmark into a small artificial-life ecosystem. Instead of one controlled ball moving to a fixed target, the tank contains multiple creatures and food resources. Creatures have mutable neural controllers, mutable morphology, energy, age, predation behavior, and mate-based reproduction. This extension is exposed through two application modes: option 5 renders the ecosystem in the MuJoCo tank for a fixed number of visible steps, while option 6 runs the same ecosystem rules headlessly for longer experiments and writes summary plots.

The ecosystem uses the same tank appearance as the navigation demo. The floor, case walls, central obstacle, translucent water volume, skybox, haze, lighting, and mouse camera controls match the earlier MuJoCo renderer. Creatures have the same initial radius as the original ball, 0.02 m, and food uses the same radius as the underwater goal marker, 0.01 m.

22 Creature State and Energy Model

Each creature stores its position, velocity, energy, age, morphology genome, and neural controller. The controller observes local motion and nearby resources and produces three thrust components plus an attack intent. Motion is evaluated in the fluid volume with water current, drag, mass, motor force, and boundary handling. The ecosystem prevents both creatures and food from spawning inside the central obstacle, and creatures that enter the obstacle are projected back out with an energy penalty and a bounce impulse.

Energy is the central survival variable:

$$E_{t+1} = E_t + E_{\text{food}} + E_{\text{prey}} - E_{\text{metabolism}} - E_{\text{motion}} - E_{\text{age}}.$$

Food increases energy, motor effort decreases it, and old age introduces an additional metabolic penalty after a configured age threshold. A creature dies from starvation when energy reaches zero, from old age when it exceeds its maximum age, or from predation when successfully eaten by another creature. The simulation reports these causes separately as starvation, old-age, and predation deaths.

23 Guided Behavior

Purely random neural controllers tended to drift into walls or fail to find food and mates. To make the ecosystem resemble the waypoint-following behavior of option 1, option 5 and option 6 blend the mutable neural controller with a guided steering term. Hungry creatures steer toward the nearest food. Creatures above the reproduction energy threshold steer toward the nearest eligible mate. Hungry, sufficiently aggressive creatures may instead steer toward weaker prey and set their attack intent. A wall and obstacle avoidance term pushes the creature away from tank boundaries and the central block.

The neural controller is still present and still mutates, but the guidance term keeps early generations from collapsing before selection has useful behavior to act on. In the current interactive preset, guidance contributes 90% of the action and the neural controller contributes the remaining 10%.

24 Reproduction and Mutation

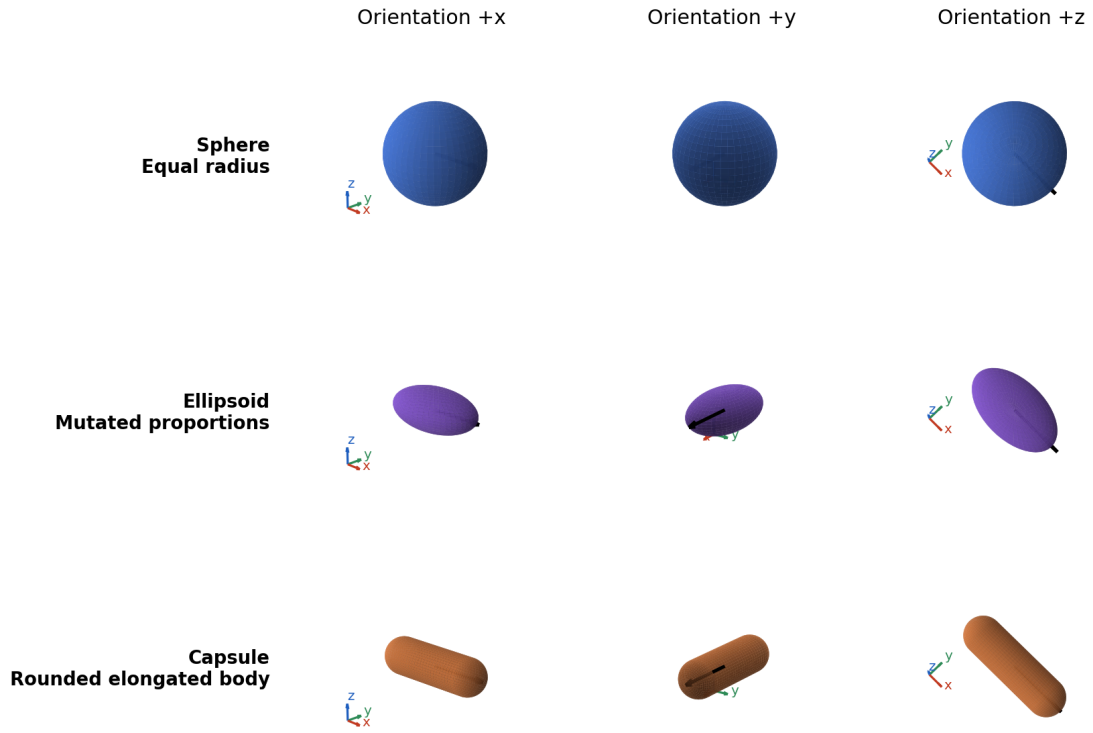
Reproduction now requires two nearby eligible creatures rather than a single creature cloning itself. Both parents must exceed the reproduction energy threshold and must be within the mating distance. When reproduction occurs, both parents pay a mating energy cost and a child is spawned near the midpoint between them.

The child controller is produced by crossover between the two parent neural networks followed by parameter mutation. The child morphology is produced by averaging parent traits and then applying mutation. Shape mutation is reward-guided rather than accepting a purely random mutation. For each birth, the simulator creates multiple morphology candidates, including explicit sphere, ellipsoid, and capsule alternatives. These candidates are scored by a morphology reward that favors efficient thrust-to-mass ratio, lower drag area, useful sensor range, defense, and balanced aggression. The highest-scoring candidate becomes the child.

The bounding radius remains fixed at 0.02 m, so mutation changes the body’s proportions and type rather than simply making larger creatures. Rendered creatures can therefore appear as spheres, ellipsoids, or capsules. Each mutated body also stores a random orientation quaternion, and MuJoCo uses that orientation when drawing the shape.

The hydrodynamic drag model now depends on this shape and orientation. Spheres use an orientation-invariant projected area and drag coefficient. Ellipsoids use the projected area of the rotated ellipsoid along the local flow direction, so broadside orientations experience higher drag than streamlined orientations. Capsules interpolate between a low-drag nose-first area and a higher-drag side-on area. This makes the energy ranking sensitive to the actual evolved shape-orientation combination rather than treating all creatures as equivalent spheres in the fluid.

Evolved Creature Shape Types and Orientation Buckets



Black arrow marks the local forward axis used for orientation grouping; colored mini-axes show world x/y/z.

Figure 17: Reference table for the three rendered morphology classes used by the ecosystem. All shapes keep the same maximum bounding radius as the original ball while allowing the proportions and orientation to evolve. Columns show representative orientation buckets; the black arrow indicates the creature’s local forward axis used for orientation grouping.

25 Predators and Old Age

Predation is enabled but gated to avoid immediate population collapse. A creature only hunts when its aggression exceeds the predator threshold and its energy is below a hunger threshold. The attack must also exceed the attack intent threshold and the prey must be within biting range. A successful predator receives a fraction of the prey’s energy, while the prey is removed from the population.

Old age adds a second long-term population pressure. After an old-age start threshold, creatures pay an increasing age-related energy cost. At the maximum age, they are removed and counted as old-age deaths. This prevents very successful early creatures from occupying the population indefinitely and gives new mutations a chance to enter the ecosystem.

26 Long-Run Output

Option 6 runs the same ecosystem rules as option 5 without opening a MuJoCo window. It repeats the long simulation across multiple random seeds and can execute those seeds in parallel worker processes. The default worker selection is automatic and caps at six workers. As each seed finishes, the command prints progress with the completed seed count, elapsed time, final population, births, deaths, and food eaten. It then prints each seed's final summary, including population, capacity, food count, generation, step births, total births, capacity-blocked births, deaths, old-age deaths, predation deaths, mean energy, food eaten, and prey eaten. At the end of the repeated run, it saves a final-seed energy analysis plot to

```
report/images/ecosystem/energy_by_shape_orientation.png.
```

It also saves an aggregate repeated-seed ranking to

```
report/images/ecosystem/aggregate_energy_ranking.png.
```

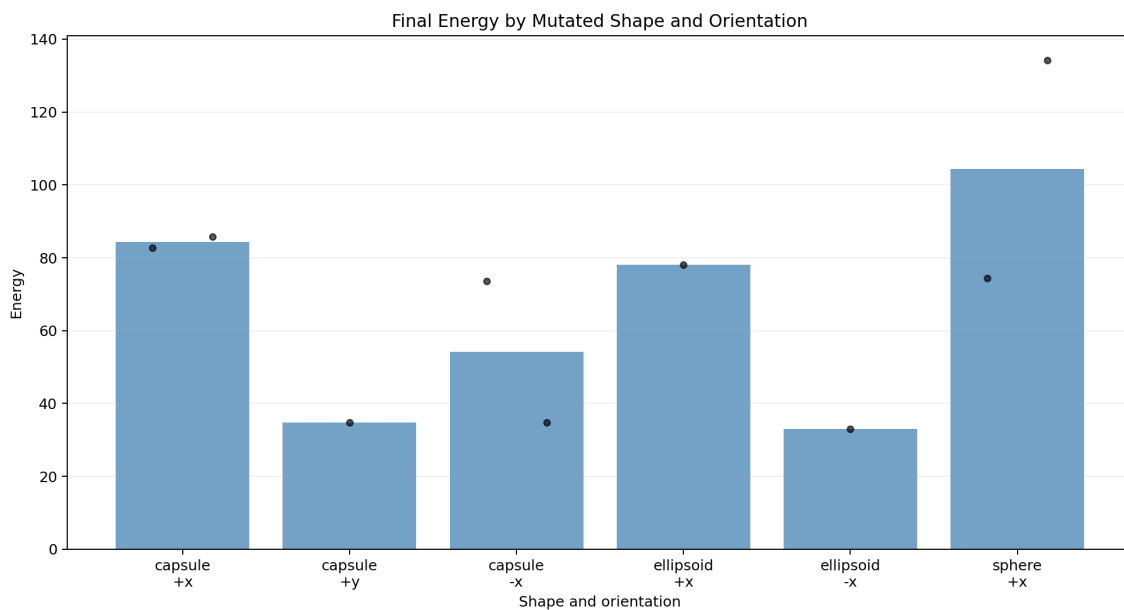


Figure 18: Final creature energy grouped by evolved shape and orientation bucket. Blue bars show group mean energy; black dots show individual creatures.

The final-seed plot groups living creatures by shape and by the world-axis bucket corresponding to the body's local forward direction. To estimate a more general ranking, option 6 aggregates all final creature energies from the repeated seeded runs and ranks shape-orientation groups by mean final energy.

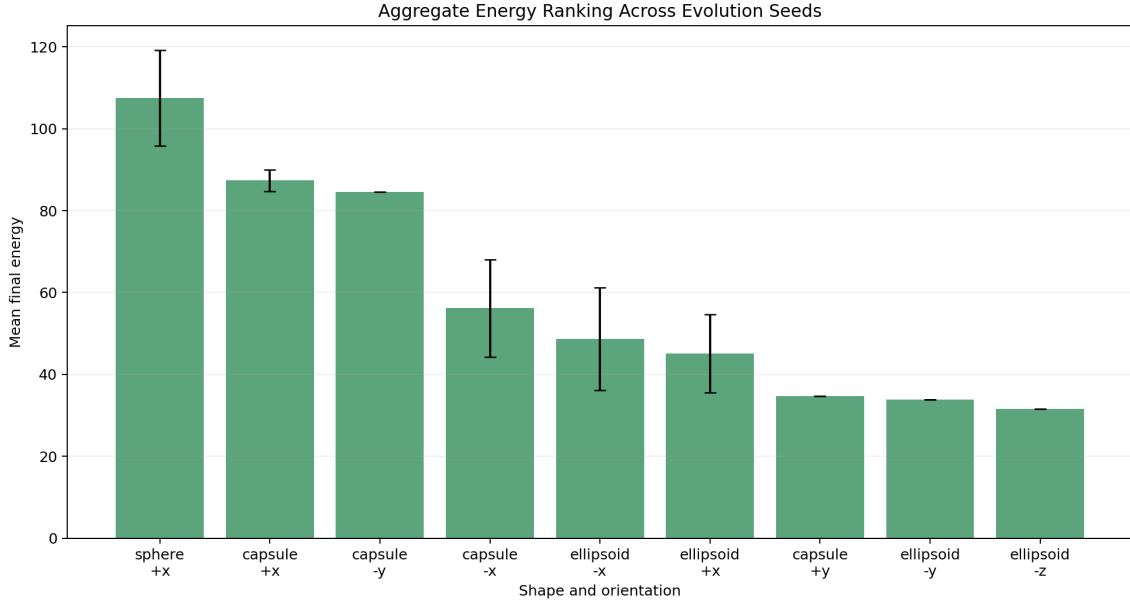


Figure 19: Aggregate shape-orientation energy ranking across repeated evolutionary simulations. Bars show mean final energy across all living creatures in the repeated-seed runs; error bars show standard error.

This aggregate plot is the current approximation of a universal ranking for the chosen ecosystem settings. It is still conditional on food density, guidance strength, predation rules, old-age parameters, obstacle layout, and number of seeds, but it is more robust than a single-run summary because the ranked groups are pooled across independent evolutionary histories.

27 Ecosystem Extension Conclusion

The ecosystem extension adds an evolutionary layer on top of the MuJoCo and fluid tank infrastructure. Creatures now search for food, seek mates, reproduce through neural and morphological crossover, mutate into different body shapes, age, die, and in some cases become predators. The implementation remains a hybrid model: MuJoCo renders the tank and creature bodies, while the ecosystem module owns energy, birth, death, selection, and behavior rules. The Navier–Stokes field supplies the evolving fluid environment used by the rendered ecosystem mode.

Future work should run repeated long simulations across seeds, export lineage trees, compare guided and unguided evolution, and quantify whether particular shape-orientation groups reliably outperform others under the same food, predation, and old-age settings.